

Code for ‘Targeted Maximum Likelihood Estimation of Vaccine Effectiveness and Immune Correlates in Test-Negative Design Studies with Missing Data’

Application 2: Inference on Immune Markers as Exposure-Proximal Correlates of COVID-19

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This R markdown provides code to perform the analyses from ‘Application 2: Inference on Immune Markers as Exposure-Proximal Correlates of COVID-19’ in the manuscript “Targeted maximum likelihood estimation of vaccine effectiveness and immune correlates in test-negative design studies with missing data” by L.I.B. Andrews, L. van der Laan, and P. Gilbert. This manuscript introduces a targeted maximum likelihood estimation approach that can be used to study the association between immune markers present around the time of potential pathogen exposure and a clinical disease endpoint using test-negative design (TND) study. We demonstrate our approach on a two-phase TND immune correlates study cohort sampled from the Moderna Coronavirus Efficacy (COVE) phase 3 randomized placebo-controlled trial (RCT) (El Sahly et al. 2021).

The COVE RCT data cannot be shared so instead, we generate a toy RCT dataset to illustrate how we conducted our data application in the manuscript. To obtain our phase one TND study cohort, we apply participant-based sampling with censoring for COVID-19 to the participants who obtained SARS-CoV-2 testing and met the symptom definition during the RCT (De Serres et al. 2013; Andrews et al. 2025). To select the participants that should have their immune markers measured, we sample without replacement to attain a 1:3 case-noncase ratio within each stratum defined by geographic region. To obtain the phase two TND study cohort, we exclude participants from the sampled subset who do not have their immune markers successfully measured or have indications of an anamnestic response from SARS-CoV-2 infection. Using the TND study cohort, we calculate our targeted maximum likelihood estimator (TMLE), a pseudo-likelihood logistic regression estimator, and an ordinary logistic regression estimator of the causal conditional risk ratio of COVID-19 when comparing vaccinated healthcare-seeking SARS-CoV-2 naïve individuals with the same characteristics who have immune marker levels greater than or equal to vs. below a given threshold.

Generating Data

Since we cannot share the COVE data, we generate a toy phase 3 COVID-19 RCT cohort and identify participants who would have been eligible for a TND immune correlates study from this RCT cohort. This data-generating code has been adapted from `Evaluation of TND with RCTs.Rmd`, which provides code for the Andrews et al. 2025 article, “Evaluating the Test-Negative Design for COVID-19 Vaccine Effectiveness Using Randomized Trial Data: A Secondary Cross-Protocol Analysis of 5 Randomized Clinical Trials” (<https://github.com/leahandrews/eval-tnd-w-rcts>). Data-generating code for the phase one TND cohort also closely follows `COVID-19 Vaccine Effectiveness Application.rmd`, which provides code for ‘Application 1: Inference on COVID-19 Vaccine Effectiveness’ in our TMLE manuscript (<https://github.com/leahandrews/semiparametric-tmle-tnd>).

RCT Cohort

Our toy RCT dataset consists of 30,000 SARS-CoV-2 naïve participants who were randomly assigned to receive one dose of the COVID-19 vaccine or placebo at a 1:1 ratio within three randomization strata. The blinded phase of the RCT takes place from August 1, 2020 to March 31, 2021. For simplicity, all participants were vaccinated in the first two weeks of August and no participants were unblinded early or lost to follow-up. At RCT enrollment, age, United States (US) Census region, presence of any comorbidities, and risk score of acquiring COVID-19 were measured for all participants. There are no missing data on any covariates.

```
## Demographic and Clinical Characteristics for RCT Cohort
# One Row per Participant
set.seed(7)
nsubj <- 30000

demo.df00 <- data.frame(USUBJID = paste0("ID", 1:nsubj),

  AGE = round(runif(nsubj, min = 18, max = 80)),

  # US Census Regions
  REGION = sample(c("Midwest", "NE", "South", "West"), nsubj,
    replace = TRUE, prob = c(2/10, 1/10, 5/10, 2/10)),

  # Presence of Comorbidities
  COMORB = sample(c("Comorbidities", "None"), nsubj,
    replace = TRUE, prob = c(1/4, 3/4)),

  # Risk Score of Acquiring COVID-19
  risk_score = rnorm(nsubj, mean = -3, sd = 0.5),

  # Intervention Date (Placebo or Vaccine)
  DOSEDT = sample(seq(as.Date('2020/08/01'),
    as.Date('2020/08/14'), by="day"),
    nsubj, replace = TRUE),

  # End of Blinded Phase Date
  ENDDT = as.Date('2021/03/31')) %>% mutate(

  # Randomization Strata
  STRATA = case_when(
    AGE < 65 & COMORB == "Comorbidities" ~ "< 65 and Comorb",
    AGE < 65 & COMORB == "None" ~ "< 65 and No Comorb",
    AGE >= 65 ~ ">= 65")
)

demo.df0 <- demo.df00 %>% group_by(STRATA) %>%
  mutate(#1:1 Vaccine vs. Placebo by Randomization Strata
    VACCINE = sample(rep(0:1, length.out = n())) %>% ungroup()
```

During the RCT, participants may experience an illness episode or symptomatic period that prompts them to obtain SARS-CoV-2 testing. At the time of the first symptom-triggered SARS-CoV-2 test, participants also have their blood drawn and stored for later immunological assessment. In this dataset, participants can experience 0-5 illness episodes during the study and the date of any symptom onset (ONSETDT) and first date of meeting the COVID-19 symptom definition (SYMDT) are documented. By definition, the date of meeting the symptom definition must occur on or after date of symptom onset. Illness episodes may be

truly caused by SARS-CoV-2 or by a non-SARS-CoV-2 illness (e.g., other respiratory pathogens, allergies). By construction, the COVID-19 vaccine protects against COVID-19 (vaccine efficacy = 80%) and does not affect other causes of COVID-19-like symptoms.

Participants obtain 1-10 SARS-CoV-2 diagnostic tests before and/or during an illness episode. While an illness episode may be truly caused by SARS-CoV-2 (`eps_sars_inf=1`), our SARS-CoV-2 test results (POSTEST) are subject to misclassification (100% specificity, 85% sensitivity). Thus, the unobserved true SARS-CoV-2 infection status of an illness episode, `eps_sars_inf`, does not always match the observed SARS-CoV-2 test results from an illness episode, POSTEST.

```
## Adding Illness Episode Information
demo.df <- demo.df0 %>% mutate(
  # True SARS-CoV-2 Infection Status at Any Point during Blinded Phase
  # (Unknown in Observed Clinical Data)
  # Vaccine Protects Against SARS-CoV-2 Infection
  any_sars_inf = rbinom(nrow(.), 1, exp( log(0.04) + log(0.2)*VACCINE) ),

  # Number of COVID-19 Illness Episodes
  n_sars_eps = any_sars_inf*sample(1:2, nsubj, replace = TRUE, prob = c(0.95, 0.05)),

  # Number of Non-COVID-19 Illness Episodes
  n_nonsars_eps = sample(0:3, nsubj, replace = TRUE, prob = c(0.9, 0.07, 0.02, 0.01)))

## Illness Episode Characteristics
# Creating One Row Per Illness Episode

# COVID-19 Illness Episodes
sars.eps.df0 <- uncount(demo.df , n_sars_eps) %>%
  # Illness Episode SARS-CoV-2 Infection Status
  mutate(eps_sars_inf = 1) %>% select(-n_nonsars_eps)

# Non-COVID-19 Illness Episodes
non.sars.eps.df0 <- uncount(demo.df , n_nonsars_eps) %>%
  # Illness Episode SARS-CoV-2 Infection Status
  mutate(eps_sars_inf = 0) %>% select(-n_sars_eps)

# Adding Testing and Symptom Information for Both Types of Illness Episodes
eps.df00 <- rbind(sars.eps.df0, non.sars.eps.df0) %>% arrange(USUBJID)
eps.df0 <- eps.df00 %>% mutate(
  # Number of SARS-CoV-2 Tests within an Illness Episode
  ntests = sample(1:10, nrow(.), replace = TRUE),

  # Date of Symptom Onset Beginning of Illness Episode)
  ONSETDT = sample(seq(as.Date('2020/09/01'), as.Date('2021/03/31'),
    by="day"), nrow(.), replace = TRUE),

  # Date of Meeting the COVID-19 Symptom Definition
  # For Simplicity, All Our Illness Episodes Meet the
  # Symptom Definition But This May Not be True in Other Datasets
  SYMDT = ONSETDT + sample(0:5, n(), replace = TRUE))%>%
  arrange(USUBJID, ONSETDT)

eps.df <- eps.df0 %>% group_by(USUBJID) %>% arrange(ONSETDT) %>%
  # Label Illness Episodes in Chronological Order
```

```

mutate(EPIISODE = 1:n()) %>% arrange(USUBJID)

## Testing Characteristics
# Creating One Row per SARS-CoV-2 Test Result
test.df0 <- uncount(eps.df, ntests)
test.df <- test.df0 %>% group_by(USUBJID)%>% mutate(
  # Test Date
  TESTDT = ONSETDT + sample(-3:15, n(), replace = TRUE),
  # Test Result Roughly Based on SARS-CoV-2 PCR Sensitivity and Specificity
  # 1 = positive, 0 = negative
  POSTEST = eps_sars_inf*rbinom(n(), 1, 0.85)) %>%
  # Only Include Tests while Blinded
  filter(TESTDT <= as.Date('2021/03/31'))

```

We use the demographic and testing data to identify RCT COVID-19 cases (RCTCASE=1), or participants who meet the COVID-19 symptom definition and obtain a SARS-CoV-2 positive test at least 14 days after vaccination. For this RCT, these criteria can be met in either order as long as both dates are within 15 days apart. The remaining RCT participants were right-censored at the end of the blinded phase.

```

## Creating RCT Analysis Dataset

# Identify First SARS-CoV-2 Positive Test from RCT COVID-19 Cases
# (In this Example Dataset, All Illness Episodes Met the Symptom Definition)
rct.case.df <- test.df %>%
  # Only take those who test SARS-CoV-2 Positive
  filter(POSTEST == 1) %>% group_by(USUBJID)%>%
  slice_min(TESTDT, with_ties = FALSE)

# Adding COVID-19 Cases' Testing Information to Demographic Information
rct.df0<- left_join(demo.df, rct.case.df[c("USUBJID", "SYMDT", "TESTDT", "POSTEST")])
rct.df <- rct.df0 %>% rowwise() %>% mutate(
  # Endpoint Date: Unblinding Date or
  # Earliest Date of Meeting Symptom Definition and Positive Test
  RCTADT = if_else(is.na(POSTEST), ENDDT, min(SYMDT, TESTDT))) %>%
  ungroup() %>% mutate(
  # RCT COVID-19 Case Status
  RCTCASE= ifelse(is.na(POSTEST), 0, 1),
  # Time to COVID-19 or Censoring
  RCTTT = as.numeric(RCTADT - (DOSED + 14) + 1))

# RCT Cases
rct_case_ids<- subset(rct.df, RCTCASE == 1)$USUBJID

with(rct.df, table(VACCINE, RCTCASE))

```

```

##          RCTCASE
## VACCINE      0      1
##          0 14411   590
##          1 14874   125

```

From the RCT cohort of 30,000 participants, 715 participants were COVID-19 cases.

Two Phase TND Immune Correlates Study

Phase One TND Study Cohort

To create our phase one TND study cohort from the RCT data, we retain participants who received the vaccine and obtained eligible SARS-CoV-2 tests, which must occur at least 14 days after vaccination, within ten days after symptom onset, after meeting the symptom definition, and while blinded. We restrict to participants who received the vaccine because our study population consists of individuals who are naïve to SARS-CoV-2 and would not have been able to generate an immune response to SARS-CoV-2 at potential exposure. Additionally, while our application in the manuscript restricted to participants who obtained SARS-CoV-2 testing within ten days of symptom onset, TND immune correlate studies analyzing neutralizing and/or binding antibody levels may consider restricting SARS-CoV-2 tests within five or seven days after symptom onset to reduce the chance of measuring immune marker levels that have been perturbed from recent SARS-CoV-2 infection in TND cases (Sumner et al. 2024; Painter et al. 2023).

In the TND analysis, we define illness episodes as symptomatic periods that last from symptom onset to at least 15 days after symptom onset and are separated by eligible SARS-CoV-2 tests that occur at least 30 days apart. We propose distinguishing illness episodes by eligible SARS-CoV-2 test dates rather than dates of symptom onset and resolution because symptom dates would be more difficult to recall and collect in a TND study. Since the illness episode variable EPISODE is defined differently in the RCT, we apply a stricter criteria to the existing RCT illness episode EPISODE variable. We define a positive episode as an illness episode that triggers at least one eligible SARS-CoV-2 positive test and a negative episode as an illness episode that does not trigger any SARS-CoV-2 positive tests and only triggers eligible SARS-CoV-2 negative tests.

```
# Only Retain TND Eligible Tests from Vaccinated Individuals
tnd.test.df <- test.df %>% filter(# Received Vaccine
                                VACCINE == 1 &
                                # Test At Least 14 Days After Vaccination
                                TESTDT >= (DOSEDT+14) &
                                # Test within 10 Days After Symptom Onset
                                TESTDT >= ONSETDT &
                                TESTDT <= (ONSETDT +10) &
                                # Test After Meeting the Symptom Definition
                                TESTDT >= SYMDT &
                                # Test While Blinded
                                TESTDT <= ENDDT) %>%
  mutate(# Quantitative Calendar Time from Sept 1, 2020 Reference Date
         # (for TMLE Approach)
         CALTIME = as.numeric(TESTDT - as.Date("2020-09-01")))

# Function that Identifies Tests That are at Least 30 days from Each Other
# Must be Applied to Data Grouped by Participant ID and Sorted by Testing Date
# Inputs:
#   col_vals: Values from a Date or Numeric Column
# Outputs: Boolean Vector, TRUE meaning rows are at least 30 days apart
#   First Value will Always be True
#
over30_func <- function(col_vals){
  # MUST BE USED AFTER GROUPED BY USUBJID AND SORTED BY DATE
```

```

# Starts with TRUE Because Have to Keep First Test
empty_vec <- c(TRUE)
nobs <- length(col_vals)
if(nobs > 1){
  for(i in 1:(nobs - 1)){
    # Are Two Tests at Least 30 Days Apart?
    val <- col_vals[i + 1] - col_vals[i] >= 30
    empty_vec <- c(empty_vec, val)
  }
}
empty_vec
}

# Take One Test per RCT-Defined Illness Episode
# (Earliest Negative Test if All Negative Tests, or Earliest Positive Test)
spec.df0 <- tnd.test.df %>% group_by(USUBJID, EPISODE)%>%
  arrange(USUBJID, TESTDT) %>% slice(which.max(POSTEST))

# Only Retain Illness Episodes That Are at Least 30 Days Apart
spec.df01 <- spec.df0 %>% group_by(USUBJID) %>%
  arrange(TESTDT)%>% mutate(SEPILL = over30_func(TESTDT))
spec.df02 <- spec.df01 %>% filter(SEPILL==TRUE)

# Only Retain Earliest Positive Episode
spec.df <- spec.df02[!(duplicated(spec.df02[c("USUBJID", "POSTEST")])) &
  spec.df02$POSTEST==1), ]

```

To obtain our phase one TND study cohort, we apply participant-based sampling with censoring for COVID-19. Under this TND sampling method, cases are participants with at least one positive episode and contribute their earliest eligible SARS-CoV-2 positive test from their first positive episode. Noncases are participants with no positive episodes and only negative episodes and contribute their earliest eligible SARS-CoV-2 negative test from their first negative episode.

```

## Constructing Phase One TND Study Cohort

# Participant-Based Sampling with Censoring
phase1.df0 <- spec.df %>% group_by(USUBJID) %>% arrange(USUBJID, TESTDT) %>%
  slice(which.max(POSTEST))

# Tertiles in Phase 1 TND Participants
caltimetert1 <- quantile(phase1.df0$CALTIME, probs = c(0, 1/3, 2/3, 1))

phase1.df <- phase1.df0 %>%
  mutate(# Calendar Time in Phase One Tertiles
    # (to Account for Smaller Cells in Standard TND Analyses)
    CALTIMEp1 = cut(CALTIME, caltimetert1, include.lowest = TRUE),

    # Two Week Calendar Time
    # (Many TND Analyses Adjust for Categorical Calendar Time)
    TWOWEEK = cut(TESTDT, "2 weeks"),

    ## Phase Two Variables (to be Updated)

```

```

# Indicator if Selected to Have Immune Markers Measured
DImm = 0,

# Indicator if Participants Selected to Have Immune Markers
# Measured are Missing Immune Markers
missmarker = NA,

# Immune Marker Level
marker = NA,

# Indicator if Case is Anti-N Seropositive
nseropos = NA,

# Indicator if in Phase Two TND Study Cohort
D = 0)

with(phase1.df, table(VACCINE, POSTEST))

```

```

##          POSTEST
## VACCINE      0      1
##          1 1323  108

```

Our phase one TND study cohort has 1431 participants and 108 cases.

Phase Two TND Study Cohort

Now that we've identified the RCT participants who would have been eligible for a TND immune correlates study (i.e., phase one TND study cohort), must select a subset of participants to measure immune markers from their stored specimen. We plan to measure immune markers from all cases and a subset of noncases sampled without replacement from the phase one TND cohort to attain a 1:3 case-noncase ratio within each stratum defined by US geographic region (Midwest, Northeast, South, and West).

The phase two TND study cohort may be smaller than the selected subset because we exclude participants in the selected subset whose immune markers were not successfully measured (for reasons other than two-phase sampling design) or who were anti-nucleocapsid seropositive, which may indicate that their immune marker levels have been affected by an anamnestic response from SARS-CoV-2 infection (Follmann et al. 2025).

```

## Selecting Participants to Measure Immune Markers
# All Cases and Three Times as Many Noncases in Each Region Stratum

# Number of Cases in Each Region Stratum
nregioncases <- with(phase1.df, table(REGION, POSTEST))

# Midwest
noncase_mw_ids <- subset(phase1.df, POSTEST == 0 & REGION == "Midwest")$USUBJID
marker_noncase_mw_ids <- sample(noncase_mw_ids,
                                nregioncases["Midwest", "1"] * 3,
                                replace = FALSE)

# NE
noncase_ne_ids <- subset(phase1.df, POSTEST == 0 & REGION == "NE")$USUBJID
marker_noncase_ne_ids <- sample(noncase_ne_ids,
                                nregioncases["NE", "1"] * 3,

```

```

        replace = FALSE)

# South
noncase_s_ids <- subset(phase1.df, POSTEST == 0 & REGION == "South")$USUBJID
marker_noncase_s_ids<- sample(noncase_s_ids,
                              nregioncases["South", "1"] * 3,
                              replace = FALSE)

# West
noncase_w_ids <- subset(phase1.df, POSTEST == 0 & REGION == "West")$USUBJID
marker_noncase_w_ids<- sample(noncase_w_ids,
                              nregioncases["West", "1"] * 3,
                              replace = FALSE)

## Participants who were Selected for Phase Two TND Study Cohort
phase1.df <- phase1.df %>% mutate(

  # Identifying Participants who were Selected to have Immune Markers Measured
  DImm = ifelse(POSTEST == 1 , 1, DImm),
  DImm = ifelse(USUBJID %in% c(marker_noncase_mw_ids,
                              marker_noncase_ne_ids,
                              marker_noncase_s_ids,
                              marker_noncase_w_ids), 1, DImm),

  # Indicator if Participants Selected to Have Immune Markers
  # Measured are Missing Immune Markers (e.g., Assay Issues, Specimen not Stored Properly)
  missmarker = ifelse(DImm == 1, sample(0:1, 1, prob = c(0.99, 0.01)), missmarker),

  # Immune Marker Level is Lower for Cases
  marker = ifelse(missmarker == 0 , rexp(1, rate = 1/(200 - 100*POSTEST)), NA),

  # Indicator if Case is Anti-N Seropositive
  nseropos = ifelse(DImm ==1,
                    ifelse( POSTEST == 1,
                           sample(0:1, 1, prob = c(0.98, 0.02)), 0),
                    nseropos),

  # D for Delta in Manuscript (i.e., Phase Two TND Study Cohort)
  # Removing Those who are Anti-Nucleocapsid Seropositive or
  # Missing Immune Markers (Not from Two-Phase Sampling)
  D = ifelse(DImm == 1 & nseropos == 0 & missmarker == 0, 1, D)
)

phase2.df <- phase1.df %>% filter(D == 1)

# Table of Cases and Noncases in Phase Two TND Study Cohort by Region Stratum
with(phase2.df, table(REGION, POSTEST))

```

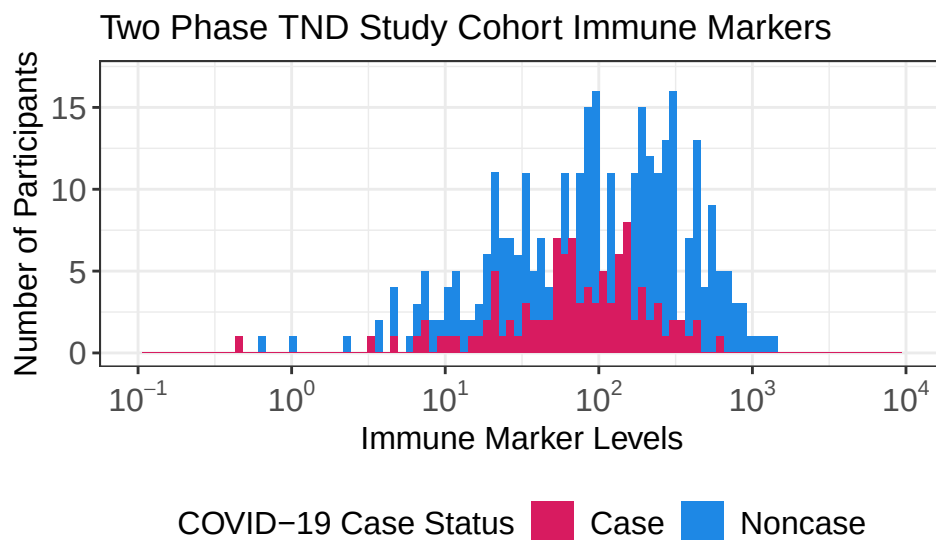
```

##          POSTEST
## REGION      0    1
## Midwest    72   24
## NE         27    7

```

```
## South 158 51
## West 64 21
```

```
phase2.df %>% ggplot( aes(log(marker, 10), fill = factor(POSTEST))) +
  geom_histogram( bins = 100)+ theme_bw() + ylim(0, 17) +
  labs(y = "Number of Participants", x = "Immune Marker Levels",
       fill = "COVID-19 Case Status",
       title = "Two Phase TND Study Cohort Immune Markers") +
  scale_fill_manual(labels = c("Noncase", "Case"),
                    values = c("0" = "#1E88E5", "1" = "#D81B60"),
                    guide = guide_legend(reverse = TRUE))+
  scale_x_continuous(limits = c(-1, 4), breaks = -2:4,
                    labels = parse(text = paste0("10^", -2:4)))+
  theme(axis.title.x = element_text( size = 12),
        axis.title.y = element_text(size = 12),
        axis.text.x = element_text(size = 12),
        axis.text.y = element_text(size = 12),
        legend.title = element_text(size = 12),
        legend.text = element_text(size = 12),
        legend.position = "bottom")
```



Currently, our TMLE approach is restricted to accommodating binary exposure variables. Thus, we investigate immune marker levels dichotomized by a threshold to study if some thresholds are associated with a greater demarcation in COVID-19 risk. We assessed thresholds (greater than or equal to vs. below) defined from the 20th, 30th, 40th, 50th, 60th, 70th, and 80th percentile for the immune marker of interest.

```
# Quantiles based on people in phase 2 TND cohort
marker_quant<- quantile(phase2.df$marker,
                        probs = seq(0, 1, by = .1))

round(marker_quant[3:9])
```

```
## 20% 30% 40% 50% 60% 70% 80%
## 34 61 86 112 149 197 297
```

```

phase2.df <- phase2.df %>% mutate(
  # Immune Marker Levels
  MARKERlog10 = log(marker, 10),
  MARKERQ = cut(marker, marker_quant, include.lowest = TRUE),
  MARKER20 = ifelse(marker >= marker_quant["20%"], 1, 0),
  MARKER30 = ifelse(marker >= marker_quant["30%"], 1, 0),
  MARKER40 = ifelse(marker >= marker_quant["40%"], 1, 0),
  MARKER50 = ifelse(marker >= marker_quant["50%"], 1, 0),
  MARKER60 = ifelse(marker >= marker_quant["60%"], 1, 0),
  MARKER70 = ifelse(marker >= marker_quant["70%"], 1, 0),
  MARKER80 = ifelse(marker >= marker_quant["80%"], 1, 0),

  # Standardize standard_risk_score by Phase 2 Mean and SD
  risk_score_s = (risk_score - mean(phase2.df$risk_score))/
    sd(phase2.df$risk_score),

  # Standardize CALTIMEDD by Phase 2 Mean and SD
  CALTIMES = (CALTIME - mean(phase2.df$CALTIME))/
    sd(phase2.df$CALTIME))

# Adding Phase 2 Covariates to the Phase 1 Dataset
phase1.df <- left_join(phase1.df, phase2.df)

```

Statistical Analyses

In this next section, we estimate the causal conditional risk ratios of COVID-19 in a SARS-CoV-2 naive healthcare-seeking population that has been vaccinated within six months using three statistical methods in the TND study cohort.

TND COVID-19 Causal Conditional Risk Ratio Using Targeted Maximum Likelihood Estimation Under Semiparametric Logistic Regression Model

We calculate a TMLE of the TND sample conditional odds ratio of COVID-19 using a partially linear logistic regression model. We use the `causalglm` R package developed by Lars van der Laan (<https://tlverse.org/causalglm/>), which implements semiparametric and nonparametric generalized linear models using targeted maximum likelihood estimation to conduct causal inference on heterogeneous treatment effects. The package relies on `tlverse/tmle3` for targeted maximum likelihood estimation and `tlverse/sl3` and `tlverse/hal9001` for machine-learning. The package is in active development so please contact Lars van der Laan if there are any issues or concerns (lvdlaan@uw.edu). Make sure to load the most up-to-date version of these packages from Github.

```

# Install devtools CRAN package
if(!require(devtools)) {
  install.packages(devtools)
}

# Install Github Packages
devtools::install_github("tlverse/causalglm")
devtools::install_github("tlverse/hal9001@master")

```

```
devtools::install_github("tlverse/tmle3@general_submodels_devel")
devtools::install_github("tlverse/sl3@Larsvanderlaan-formula_fix")
```

Since our TND immune correlates study cohort restricts to vaccinated RCT individuals, our study is subject to confounding. Thus, in our partially linear logistic regression model, we adjust for the risk score of acquiring COVID-19 (standardized), presence of comorbidities, US Census region, and quantitative calendar time of SARS-CoV-2 testing (standardized).

For estimation, we use a 10-fold cross-fitted ensemble super learner that considers a library of learners including the sample mean, logistic regression models, general additive models, multivariate adaptive regression splines, lasso regression, and ridge regression (van der Laan, Polley, and Hubbard 2007; Chernozhukov et al. 2018). The library also includes screened logistic regression and general additive model learners, which only include covariates with nonzero coefficients based on lasso regression. We assess the candidate learners using a binomial log-likelihood loss and 10-fold cross-validation and combine the nuisance function predictions using non-negative least squares regression.

```
## Ensemble Super Learner

# Define Learners without Screens
mean_lrn <- Lrnrm_mean$new()
glm_lrn <- Lrnrm_glm$new(family = binomial())
glm_intx <- Lrnrm_glm$new(formula = ~ (. )^2, family = binomial())
gam_lrn <- Lrnrm_gam$new(family = binomial())
mars_lrn <- Lrnrm_earth$new()
lasso_lrn <- Lrnrm_glmnet$new(alpha = 1, family = "binomial")
ridge_lrn <- Lrnrm_glmnet$new(alpha = 0, family = "binomial")

# Define Learners with Lasso Screen
lasso_screen <- Lrnrm_screener_coefs$new(learner =
  Lrnrm_glmnet$new(alpha = 1, family = "binomial"), threshold = 0)
glm_lasso <- Pipeline$new(lasso_screen, glm_lrn)
glm_intx_lasso <- Pipeline$new(lasso_screen, glm_intx)
gam_lasso <- Pipeline$new(lasso_screen, gam_lrn)

# Stack Learners
marker_stack <- Stack$new(
  mean_lrn,
  glm_lrn,
  glm_intx,
  gam_lrn,
  mars_lrn,
  lasso_lrn,
  ridge_lrn,
  glm_lasso,
  glm_intx_lasso,
  gam_lasso)

# SuperLearner
marker_sl <- Lrnrm_sl$new(
  learners = marker_stack,
  metalearner = Lrnrm_nnls$new(),
  loss_function = loss_loglik_binomial)
```

To run our TMLE approach for the 20th quantile, we apply the `spglm` function in `causalglm`, which currently

requires all variables in the input dataset to be numerics. We use `model.matrix` to convert all character and factor variables into numeric or indicator variables.

```
# Convert Data to Only Contain Numeric/Indicator Variables (spglm Requirement)
# Use model.matrix to Convert Character/Factor Variables for Each TND Sample
semi.model.20.df <- as.data.frame(model.matrix(~ MARKER20 +
      POSTEST + REGION + risk_score_s +
      COMORB + CALTIMES, data = phase2.df))

# Run TMLE Approach with Super Learner for Estimation
set.seed(2025)
semi.mod.20 <- spglm( ~1, semi.model.20.df,
  W = setdiff(names(semi.model.20.df),
    c("(Intercept)", "MARKER20", "POSTEST")),
  A = "POSTEST", Y = "MARKER20",
  estimand = "OR",
  sl3_Learner_Y = marker_sl,
  sl3_Learner_A = marker_sl)

# Model Output
print(semi.mod.20)
```

```
## A causalglm fit object obtained from spglm for the estimand OR with formula:
## log OR(W) = -0.183 * (Intercept)
```

```
# Example Output
summary(semi.mod.20)
```

```
## A causalglm fit object obtained from spglm for the estimand OR with formula:
## log OR(W) = -0.183 * (Intercept)
##
## Coefficient estimates and inference:
##      type      param  tmle_est      se      lower      upper  psi_exp
##   <char>    <char>    <num>    <num>    <num>    <num>    <num>
## 1:      OR (Intercept) -0.1830237 0.2774121 -0.7267413 0.360694 0.8327485
##   lower_exp upper_exp  Z_score p_value
##        <num>    <num>    <num>    <num>
## 1: 0.4834819  1.434324 0.6597538 0.50941
```

Our TMLE approach takes less than a minute using our library of learners. Applying `print` to an `spglm` object reports the object and equation. `summary` of an `spglm` object reports summary statistics for the sample log conditional odds ratio of immune marker threshold by case status, adjusting for covariates. `tmle_est` is the log conditional odds ratio and `psi_exp` is the conditional odds ratio. `lower_exp` and `upper_exp` are the exponentiated lower and upper bounds of the log conditional odds ratio 95% confidence interval.

Assuming the eleven identifying assumptions specified in our manuscript hold, we can interpret the TND sample conditional odds ratio as the causal conditional risk ratio of COVID-19 comparing vaccinated SARS-CoV-2 naïve healthcare-seeking individuals with immune marker levels greater than or equal to the twentieth quantile (34 units) to those with immune markers lower than the twentieth quantile, holding risk score of acquiring COVID-19, comorbidities, region, and calendar date constant (RR = 0.83; 95% CI: 0.48, 1.43). Using our targeted maximum likelihood estimation approach, we estimate that vaccinated SARS-CoV-2 naïve healthcare-seeking individuals with immune markers higher than 34 units have 17% lower risk of COVID-19 than those with immune markers lower than that threshold, given they share the same covariates.

Since we are interested in evaluating seven immune marker thresholds and want to visualize the results using `forestplot`, we create `semilogit_sl_tbl` so we can easily run the function in a loop and output all the desired information to create a forest plot.

```
### semilogit_sl_tbl: TMLE Approach under Semiparametric Logistic Regression Model
# with Estimation Using User-Specified Learners. Approach constructed from
# spglm function in causalglm package. Outputs list of model values.

## Function inputs:
# poi_var: string of model's exposure variable name
# e.g., "POSTEST"
# outcome_var: string of model's outcome variable name
# e.g., "MARKER20"
# model_formula: string of desired logistic regression formula
# (the function will modify the formula to match spglm formula requirements)
# e.g., "MARKER20 ~ POSTEST + REGION + COMORB + CALTIMes + risk_score_s"
# missing_var: string of indicator of observing the exposure variable name
# e.g., "D"
# learner_name: string of learner name to be included in outputted table
# e.g., "Super Learner"
# learner: sl3 Learner object to use for estimation
# df: dataset with complete information on missing_var and all variables in model_formula
# except poi_var (function excludes participants missing poi_var for analyses)
# e.g., phase1.df or phase2.df

## Function outputs:
# Data.Frame with df Information and Model Coefficients
semilogit_sl_tbl <- function(poi_var, outcome_var, model_formula,
                             missing_var = "D", learner_name,
                             learner, df){

  # Keep Individuals with Observed Exposure Variable (Delta=1)
  complete.df <- df %>% filter(get(missing_var) == 1)

  # Table of Model's Outcome Variable and Exposure Variable
  tbl <- addmargins(table(complete.df[[outcome_var]],
                        complete.df[[poi_var]]))
  proptbl <- round(prop.table(table(complete.df[[outcome_var]],
                                   complete.df[[poi_var]]), 2)*100, 0)
  tblcounts <- paste0(tbl[2,1:2], "/", tbl[3,1:2], " (", proptbl[2,], ")")

  ## Run Partially Linear Logistic Regression Model
  # To get data in right format, create dataset with only numeric and
  # binary variables with model matrix including outcome
  model_formula <- gsub(paste0(outcome_var, " *~"),
                        paste0("~", outcome_var, "+"),
                        model_formula)

  model.df <- as.data.frame(model.matrix(eval(parse(text = model_formula)),
                                         data = complete.df))

  mod <- spglm(~1, model.df,
               W = setdiff(names(model.df), c("(Intercept)", outcome_var, poi_var)),
```

```

A = poi_var, Y = outcome_var,
estimand = "OR",
sl3_Learner_Y = learner,
sl3_Learner_A = learner)

## Calculate relevant statistics
return(data.frame(StatMethod = learner_name,
  Characteristic = outcome_var,
  nPhase2 = nrow(complete.df),
  Cases = tblcounts[2],
  Noncases = tblcounts[1],
  LowCases = sum(complete.df[[poi_var]] == 1 &
    complete.df[[outcome_var]] == 0),
  HighCases = sum(complete.df[[poi_var]] == 1 &
    complete.df[[outcome_var]] == 1),
  logRR = format(round(coef(mod)$tmle_est, 2), nsmall = 2),
  logRRCIlow = format( round( coef(mod)$lower, 2), nsmall = 2),
  logRRCIhigh = format( round( coef(mod)$upper, 2), nsmall = 2),
  RRCI = paste0(format(round(coef(mod)$psi_exp, 2), nsmall = 2), " (",
    format(round(coef(mod)$lower_exp, 2), nsmall = 2), ", ",
    format(round(coef(mod)$upper_exp, 2), nsmall = 2), ")"),
  PVal = format( round(coef(mod)$p_value, 3), nsmall = 3),
  Form = model_formula,
  stringsAsFactors = FALSE)) }

# Estimate TMLE for 20th to 80th Immune Marker Quantiles
marker_vars <- paste0("MARKER", seq(20,80, by = 10))
semi.list <- vector(mode = "list", length = length(marker_vars))
set.seed(2025)

for(i in 1:length(marker_vars)){

  semi.list [[i]] <- semilogit_sl_tbl(
    poi_var = "POSTEST",
    outcome_var = marker_vars[i],
    model_formula = paste0(marker_vars[i],
      " ~ POSTEST + REGION + CALTIMES + COMORB + risk_score_s"),
    missing_var = "D",
    learner_name = "Super Learner",
    learner = marker_sl,
    phase1.df)
}

# Dataset of TMLE Results for Each Quantile
semi.tbl0 <- do.call(rbind, semi.list)

```

We run `semilogit_sl_tbl` to obtain the TMLEs of the causal conditional risk ratios of COVID-19 by the 20th, 30th, 40th, 50th, 60th, 70th, and 80th immune marker quantiles.

TND COVID-19 Causal Conditional Risk Ratio Using Pseudo-Likelihood Logistic Regression

To account for the two-phase sampling design, we estimate the TND sample conditional odds ratio of COVID-19 (which can be interpreted as a causal conditional risk ratio under the assumptions listed in the manuscript) using a pseudo-likelihood logistic regression with empirical variance estimates (Breslow and Cain 1988; Breslow and Holubkov 1997). We adjust for risk score of acquiring COVID-19, presence of comorbidities, US Census region, and tertile calendar time of SARS-CoV-2 testing and account for the strata used in the two-phase sampling design. We use the `tps` function in the `osDesign` R package to implement the pseudo-likelihood logistic regression (Haneuse, Saegusa, and Lumley 2011) and create the `pl_logit_tbl` function so we can easily run the pseudo-likelihood approach for all seven immune marker thresholds and visualize the results in a forest plot.

```
### pl_logit_tbl: Pseudo-Likelihood Logistic Regression Model with Empirical Variance
# Estimates. Approach constructed from tps function in osDesign R package.
# Outputs list of model values.

## Function inputs:
# poi_var: string of model's exposure variable name
# e.g., "MARKER20"
# outcome_var: string of model's outcome variable name
# e.g., "POSTEST"
# model_formula: string of desired logistic regression formula
# e.g., "POSTEST ~ MARKER20 + REGION + COMORB + CALTIMEp1 + risk_score"
# missing_var: string of indicator of observing the exposure variable name
# e.g., "D"
# strata_var: string of two-phase sampling strata variable name
# e.g., "REGION"
# var_type: string of "Empirical" or "Model" variance
# learner: string of method of estimation allowed from tps function
# e.g., pseudo-likelihood ("PL"), weighted likelihood ("WL"), or
# maximum likelihood ("ML")
# df: dataset with complete information on missing_var and all variables in model_formula
# except poi_var
# e.g., phase1.df

## Function outputs:
# Data.Frame with df Information and Model Coefficients

pl_logit_tbl <- function(poi_var = "A", outcome_var = "Y",
                        model_formula, missing_var = "D",
                        strata_var = "STRATA", var_type = c("Empirical", "Model"),
                        learner = "PL", df){

  # Table of STRATA and Cases
  strata_tbl <- table(df[[strata_var]], df[[outcome_var]])

  # Identify Phase Two Data
  complete.df <- df %>% filter(get(missing_var) == 1)

  # High Immune Markers in Cases and Noncases
  tbl <- addmargins(table(complete.df[[poi_var]],
                        complete.df[[outcome_var]]))
  proptbl <- round(prop.table(table(complete.df[[poi_var]],
```

```

                                complete.df[[outcome_var]]),2)*100, 0)
tblcounts <- paste0(tbl[2,1:2], "/", tbl[3,1:2], " (", proptbl[2,], ")")

## Run Pseudo-Likelihood Logistic Regression Model
mod <- tps(eval(parse(text = model_formula)), data = complete.df,
           nn0 = strata_tbl[ , "0"], nn1 = strata_tbl[ , "1"],
           group = complete.df[[strata_var]], method = learner, cohort = FALSE)

## Calculate relevant statistics
coef <- mod$coef[which(names(mod$coef) == poi_var)]
se <- ifelse( var_type == "Model",
              sqrt( mod$covm[poi_var, poi_var] ) ,
              sqrt( mod$cove[poi_var, poi_var] ) )
rr <- exp(coef)
z <- qnorm(0.975)
rrCIlow <- format(round(exp(coef - z*se), 2), nsmall = 2)
rrCIhigh <- format(round(exp(coef + z*se), 2), nsmall = 2)
rrci <- paste0(format(round(rr , 2), nsmall = 2), " (", rrCIlow, ", ", rrCIhigh, ")")
pval = format(round(2*(1 - pnorm(abs(coef/se))), 3), nsmall = 3)

## Calculating relevant statistics
return(data.frame(StatMethod = paste0(learner, " Logit ", var_type, " SE"),
                  Characteristic = poi_var,
                  nPhase2 = nrow(complete.df),
                  Cases = tblcounts[2],
                  Noncases = tblcounts[1],
                  LowCases = sum(complete.df[[outcome_var]] == 1 &
                                complete.df[[poi_var]] == 0),
                  HighCases = sum(complete.df[[outcome_var]] == 1 &
                                complete.df[[poi_var]] == 1),
                  logRR = format(round(coef, 2), nsmall = 2),
                  logRRCIlow = format(round( coef - z*se, 2), nsmall = 2),
                  logRRCIhigh = format(round( coef + z*se, 2), nsmall = 2),
                  RRCI = rrci,
                  PVal = pval,
                  Form = model_formula,
                  stringsAsFactors = FALSE)) }

```

```

# Fit Pseudo-Likelihood Logistic Regression for 20th-80th Immune Marker Quantiles
pl.tbl0 <- map_dfr(marker_vars,
                   function(i) pl_logit_tbl(poi_var = i,
                                             outcome_var = "POSTEST",
                                             model_formula = paste0("POSTEST ~ ", i,
                                                                    " + REGION + CALTIMEp1 + COMORB + risk_score"),
                                             missing_var = "D",
                                             strata_var = "REGION",
                                             var_type = "Empirical",
                                             learner = "PL",
                                             df = phase1.df))

```

We run `pl_logit_tbl` to obtain the estimated causal conditional risk ratios of COVID-19 by the 20th, 30th,

40th, 50th, 60th, 70th, and 80th immune marker quantiles using pseudo-likelihood logistic regression.

TND COVID-19 Vaccine Effectiveness Using Ordinary Logistic Regression

We also estimate the TND sample conditional odds ratio of COVID-19 using an ordinary logistic regression of COVID-19 adjusted for immune marker threshold, risk score of acquiring COVID-19, presence of comorbidities, US Census region, and tertile calendar time of SARS-CoV-2 testing. We use the `glm` function to implement the ordinary logistic regression and create the `ordlogit_tbl` function so we can easily run the method for all seven immune marker thresholds and visualize the results in a forest plot.

```
### ordlogit_tbl: Ordinary Logistic Regression Model.
# Approach implemented using glm. Outputs list of model values

## Function inputs:
# poi_var: string of exposure of interest variable name
# e.g., "MARKER20"
# outcome_var: string of outcome variable name
# e.g., "POSTEST"
# model_formula: string of desired logistic regression formula
# e.g., "POSTEST ~ MARKER20 + REGION + COMORB + CALTIMEp1 + risk_score"
# missing_var: string of indicator of observing the exposure variable name
# e.g., "D"
# df: dataset with complete information on missing_var and all variables in model_formula
# except poi_var (function excludes participants missing poi_var for analyses)
# e.g., phase1.df or phase2.df

## Function outputs:
# list that contains parameters and full glm model
ordlogit_tbl <- function(poi_var = "A", outcome_var = "Y",
                        model_formula,
                        missing_var = "D", df){

  # Only Analyze Phase 2 Data
  complete.df <- df %>% filter(get(missing_var) == 1)

  # High Immune Markers in Cases and Noncases
  tbl <- addmargins(table(complete.df[[poi_var]],
                        complete.df[[outcome_var]]))
  proptbl <- round(prop.table(table(complete.df[[poi_var]],
                        complete.df[[outcome_var]]), 2)*100, 0)
  tblcounts <- paste0(tbl[2,1:2], "/", tbl[3,1:2], " (", proptbl[2,], ")")

  ## Run Ordinary Logistic Regression Model
  mod <- glm(eval(parse(text = model_formula)),
             data = complete.df, family = "binomial")
  glm.mod <- summary(mod)

  ## Calculate relevant statistics
  coef <- glm.mod$coefficients[poi_var, "Estimate"]
  se <- glm.mod$coefficients[poi_var, "Std. Error"]
  rr <- format(round(exp(coef), 2), nsmall = 2)
  z <- qnorm(0.975)
```

```

rrCIlow <- format( round( exp(coef - z*se), 2), nsmall = 2)
rrCIhigh <- format( round( exp(coef + z*se), 2), nsmall = 2)
rrci <- paste0(rr, " (", rrCIlow, ", ", rrCIhigh, ")")
pval = format( round(glm.mod$coefficients[poi_var,"Pr(>|z|)"], 3), nsmall = 3)

return(data.frame(StatMethod = "Ordinary",
  Characteristic = poi_var,
  nPhase2 = nrow(complete.df),
  Cases = tblcounts[2],
  Noncases = tblcounts[1],
  LowCases = sum(complete.df[[outcome_var]] == 1 &
    complete.df[[poi_var]] == 0),
  HighCases = sum(complete.df[[outcome_var]] == 1 &
    complete.df[[poi_var]] == 1),
  logRR = format(round(coef, 2), nsmall = 2 ),
  logRRCIlow = format( round( coef - z*se, 2), nsmall = 2),
  logRRCIhigh = format( round( coef + z*se, 2), nsmall = 2) ,
  RRCI = rrci,
  PVal = pval,
  Form = model_formula,
  stringsAsFactors = FALSE)) }

```

```

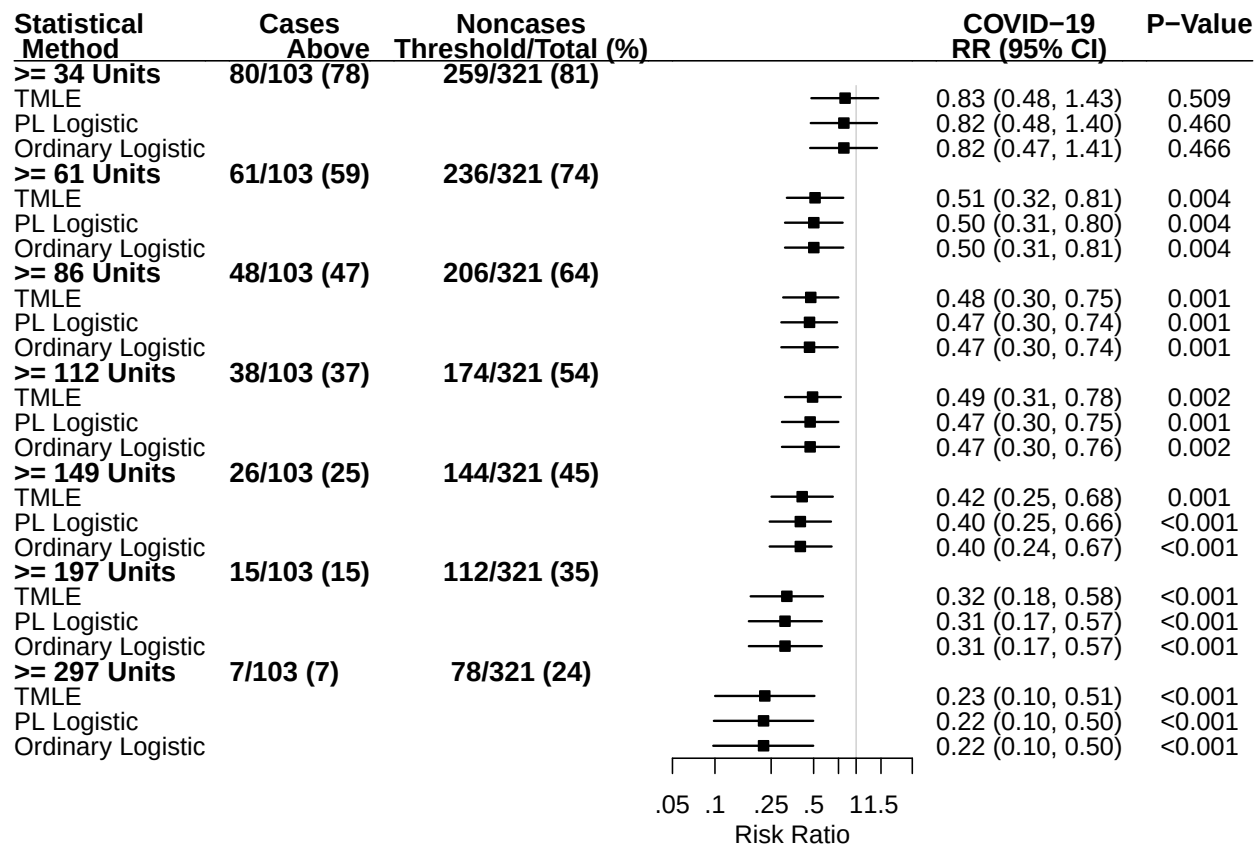
# Fit Ordinary Logistic Regression for 20th-80th Immune Marker Quantiles
ord.tbl0 <- map_dfr(marker_vars,
  function(i) ordlogit_tbl(
    poi_var = i,
    outcome_var = "POSTEST",
    model_formula = paste0("POSTEST ~ ", i,
      " + REGION + CALTIMEp1 + COMORB + risk_score"),
    missing_var = "D",
    df = phase1.df))

```

We run `ordlogit_tbl` to obtain the estimated causal conditional risk ratios of COVID-19 by the 20th, 30th, 40th, 50th, 60th, 70th, and 80th immune marker quantiles using ordinary logistic regression.

Summary of Results

We use the `forestplot` function from the `forestplot` R package to visualize the causal conditional risk ratios of COVID-19 by immune marker threshold and statistical approach.



The estimated causal conditional risk ratios of COVID-19 for all three statistical methods are highly concordant and have similar confidence intervals.

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